Katherine Deborah

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EDUCATION

University of Southern California (USC) December 2026
Master of Science in Computer Science *Los Angeles, California, USA*
Coursework: Fundamentals of Artificial Intelligence, Web Technology, Analysis of Algorithms

Government Institute of Technology June 2024
Bachelor of Technology in Information Technology *Coimbatore, India*

TECHNICAL SKILLS

Programming Language: Python, C++, Java, MATLAB, SQL, C, JavaScript, HTML, CSS
Libraries and frameworks: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, PyTorch, TensorFlow, Flask, Vue.js, React.js, Next.js, Node.js, Three.js, Streamlit
Tools and Platforms: Docker, Apache Kafka, Git, Tableau, Rest API
Databases: MySQL, PostgreSQL, MongoDB

WORK EXPERIENCE

Department of Telecommunications, Government of India Aug 2023-Dec 2023
Data Analyst *Coimbatore, India*

- Developed a full-stack web application to streamline inquiry responses using React (frontend), Flask (backend), and SQLite, integrating key insights from SIM card data to enable real-time, accurate lookups
- Automated the previously manual inquiry process, cutting response time by over 10 hours through dynamic routing and database integration
- Engineered backend algorithms to auto-route electronic files and track case status, improving departmental efficiency and transparency
- Cleaned and processed 60M+ rows of SIM card metadata (80 fields) using Python, Pandas, and Scikit-learn; conducted anomaly detection and contributed a time-saving solution to recurring data inconsistencies

PROJECTS

Video Analysis Q&A System

- Built a Streamlit-based UI for video uploads, extracting frames at 0.2 FPS and generating scene captions using the BLIP model vision-language LLM; stored captions in SQLite for efficient retrieval
- Integrated Whisper for speech-to-text transcription and Qwen LLM for natural language question answering, enabling users to query and receive context-aware responses about video content

Diabetic Retinopathy Detection

- Developed a hybrid deep learning framework combining GoogLeNet and ResNet with Adaptive Particle Swarm Optimization for enhanced feature extraction from retinal fundus images
- Classified DR severity levels using multiple ML models (Random Forest, SVM, Decision Tree, Linear Regression), achieving 94% accuracy and outperforming state-of-the-art benchmarks

Bidirectional Static Rectangle Expansion A*

- Developed an optimized bidirectional SREA* pathfinding algorithm that expands search nodes from both source and destination, reducing iterations by 50% and significantly lowering computation time without increasing memory usage
- Designed a modular architecture with map representation, search management, and real-time UI visualization, enabling lightweight deployment on mobile processors while maintaining optimal path accuracy in grid-based environments

CERTIFICATIONS

Diploma in Programming and Data Science – *IIT Madras*, Mathematics for Machine Learning - *Coursera Imperial College London,* Cloud Computing, Block Chain - *NPTEL IIT Kharagpur*